

Exam 3

ECE 2201: Circuit Analysis I

Summer 2025

July 19, 2025

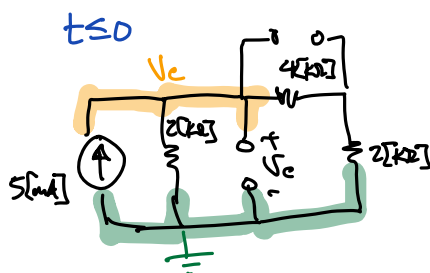
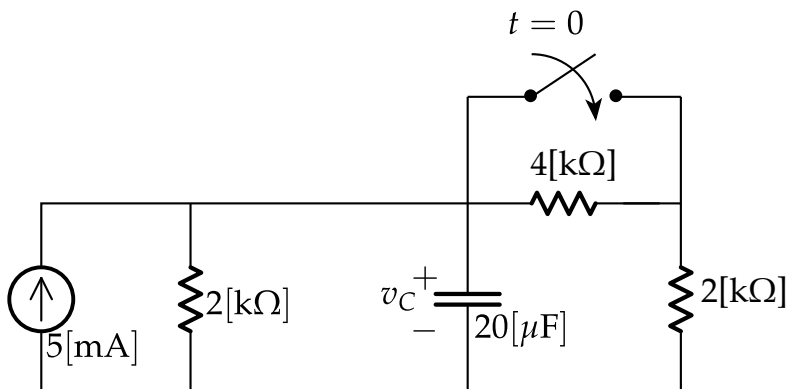
1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem and explain the steps in your solution. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 75 minutes to work on this exam.

Name: Solution ID: _____

1. _____ /25
2. _____ /25
3. _____ /25
4. _____ /25

Problem 1

The switch in the circuit below has been open for a long time before it is closed at $t = 0$. Determine $v_C(0^-)$, $v_C(\infty)$, τ , and $v_C(t)$.

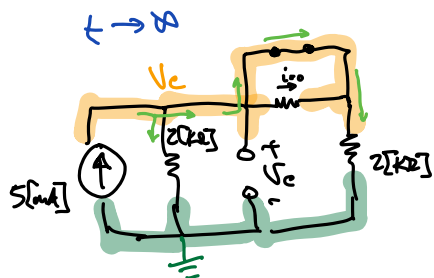


$$V_c = 5[\text{mA}] \cdot (2[\text{k}\Omega] \parallel 4[\text{k}\Omega])$$

$$\frac{8 \cdot 6^3}{8 \cdot 4^2} [\text{k}\Omega] = \frac{3}{2} [\text{k}\Omega]$$

$$V_c = 5[\text{mA}] \cdot \frac{3}{2} [\text{k}\Omega] = \frac{15}{2} [\text{V}] = 7.5 [\text{V}]$$

$$V_c(0^-) = 7.5 [\text{V}]$$



$$V_c = 5[\text{mA}] \cdot (2[\text{k}\Omega] \parallel 2[\text{k}\Omega])$$

$$= 5 [\text{V}]$$

$$V_c(\infty) = 5 [\text{V}]$$

$$R_{eq} = 2[\text{k}\Omega] \parallel 2[\text{k}\Omega] = 1[\text{k}\Omega]$$

$$\begin{aligned} \tau &= RC = 1[\text{k}\Omega] \cdot 20[\mu\text{F}] \\ &= 10^3 \cdot 20 \cdot 10^{-6} [\text{s}] \\ &= 20 [\text{ms}] \end{aligned}$$

$$\begin{aligned} v_C(t) &= v(\infty) + (v(0^-) - v(\infty))e^{-t/\tau} \\ &= (5 + (7.5 - 5)e^{-t/20[\text{ms}]}) [\text{V}] \end{aligned}$$

$$v_C(0^-) = \underline{7.5 [\text{V}]}$$

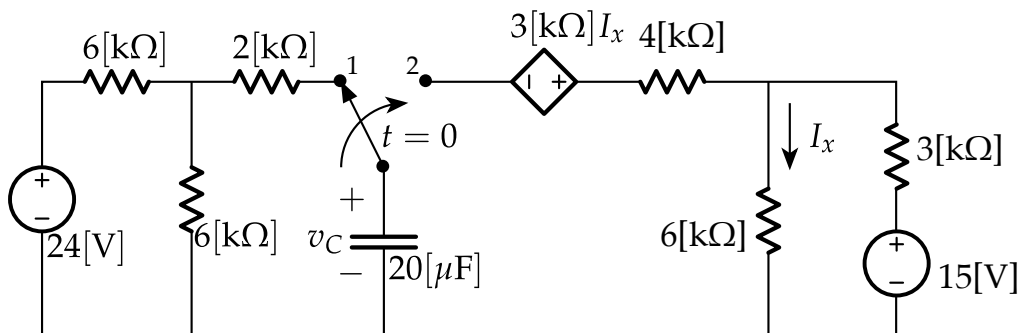
$$v_C(\infty) = \underline{5 [\text{V}]}$$

$$\tau = \underline{20 [\text{ms}]}$$

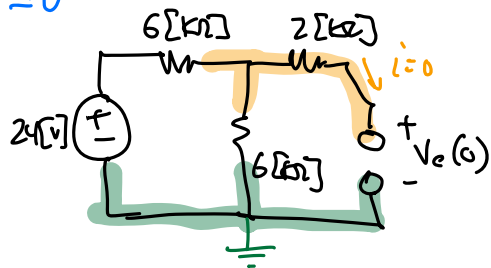
$$v_C(t) = \underline{(5 + 2.5e^{-t/20[\text{ms}]}) [\text{V}]}$$

Problem 2

The switch in the circuit below has been in position 1 for a long time before it is moved to position 2 at $t = 0$. Determine $v_C(0^-)$, $v_C(\infty)$, τ (for position 2), and $v_C(t)$.



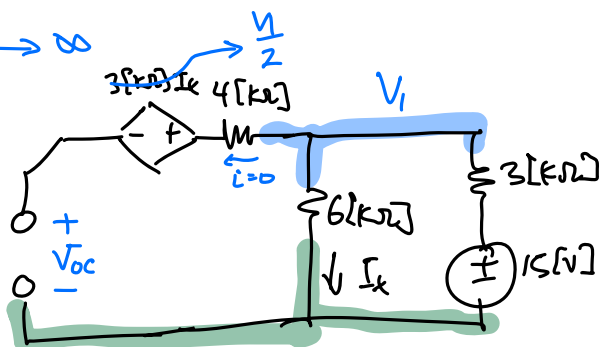
$t \leq 0$



Since $i = 0$

$$v_C(0) = 24[V] \frac{6[k\Omega]}{(6+6)[k\Omega]} = 12[V]$$

$t \rightarrow \infty$



$$I_x = \frac{V_1}{6[k\Omega]} \quad 3[k\Omega] I_x = \frac{V_1}{2}$$

$$\frac{V_1 - 15[V]}{3[k\Omega]} + \frac{V_1}{6[k\Omega]} = 0$$

or use a voltage divider:

$$\frac{6}{3+6} \cdot 15[V] = 10[V]$$

$$2V_1 - 30[V] + V_1 = 0$$

$$V_1 = 10[V]$$

$$V_{oc} = V_1 - \frac{V_1}{2} = \frac{V_1}{2} = 5[V]$$

$$R_{eq} = \frac{V_{oc}}{i_{sc}} = \frac{5[V]}{12mA} = 5[k\Omega]$$

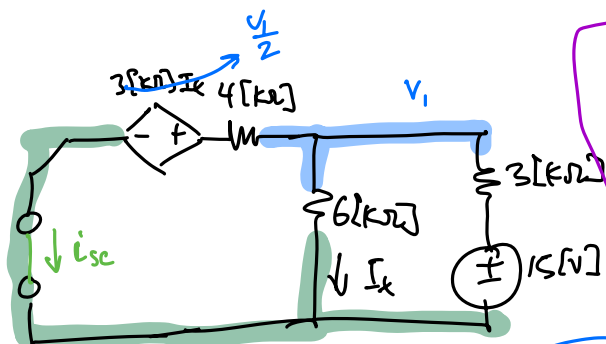
$$\tau = 5[k\Omega] \cdot 20[\mu F] = 5 \times 10^3 \cdot 20 \times 10^{-6} [s] = 0.1[s]$$

$$v_C(0^-) = 12[V]$$

$$v_C(\infty) = 5[V]$$

$$\tau = 0.1[s]$$

$$v_C(t) = (5 + 7e^{-t/0.1[s]})[V]$$



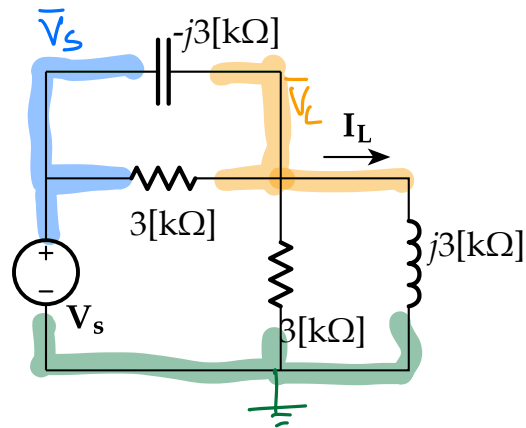
$$i_{sc} = \frac{V_1 - \frac{V_1}{2}}{4[k\Omega]} = \frac{V_1}{8[k\Omega]} = 1[mA]$$

$$\left[\frac{V_1 - \frac{V_1}{2}}{4[k\Omega]} + \frac{V_1}{6[k\Omega]} + \frac{V_1 - 15[V]}{3[k\Omega]} = 0 \right] \times 24[V]$$

$$6V_1 - 3V_1 + 4V_1 + 8V_1 - 120[V] \Rightarrow 15V_1 = 120[V] \Rightarrow V_1 = 8[V]$$

Problem 3

The circuit below has been converted to the phasor domain with $V_s = 5[V]$. Find I_L .



Node-voltage

$$\left[\frac{\bar{V}_L - \bar{V}_s}{-j3[k\Omega]} + \frac{\bar{V}_L - \bar{V}_s}{3[k\Omega]} + \frac{\bar{V}_L}{3[k\Omega]} + \frac{\bar{V}_L}{j3[k\Omega]} = 0 \right] \times 3[k\Omega]$$

$$j\bar{V}_L - j\bar{V}_s + \bar{V}_L - \bar{V}_s + \bar{V}_L - j\bar{V}_L = 0$$

$$2\bar{V}_L + (-1-j)\bar{V}_s = 0$$

$$\bar{V}_L = \frac{1+j}{2} \bar{V}_s = (2.5 + j2.5)[V]$$

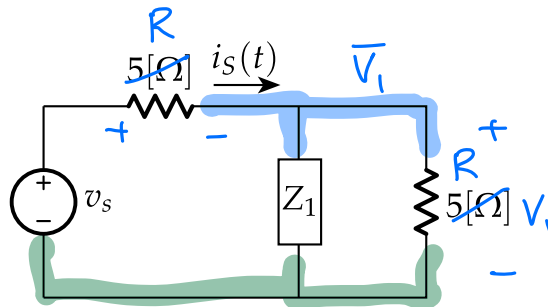
$$\bar{I}_L = \frac{\bar{V}_L}{j3[k\Omega]} = \frac{(2.5 + j2.5)[V]}{j3[k\Omega]} = \left(\frac{2.5}{3} - j\frac{2.5}{3} \right) [mA] = (0.833 - j0.833)[mA]$$

$$= 1.18[mA] \angle -45^\circ$$

$$I_L = \underline{1.18[mA] \angle -45^\circ}$$

Problem 4

- a) Find impedance Z_1 in the circuit below given that $v_s = 10 \sin(1000[\text{rad/s}]t + 60^\circ)$ [V] and $i_s = 100 \cos(1000[\text{rad/s}]t - 120^\circ)$ [mA]
 b) Sketch an equivalent model for Z_1 by identifying the components that have the equivalent impedance



KVL:

$$-\bar{V}_s + \bar{I}_s \cdot R + \bar{V}_1 = 0$$

$$V_1 = \bar{V}_s - \bar{I}_s R$$

$$\bar{V}_s = 10[\text{V}] / \underline{60^\circ - 90^\circ} = 10[\text{V}] / \underline{-30^\circ}$$

$$\bar{I}_s = 0.1[\text{A}] / \underline{-120^\circ}$$

this problem does not have a real solution, which was not my intention. will be used as bonus points.

Node-Voltage:

$$\frac{\bar{V}_1 - \bar{V}_s}{R} + \frac{\bar{V}_1}{R} + \frac{\bar{V}_1}{Z_1} = 0$$

$$\frac{\bar{V}_1}{Z_1} = -\frac{2\bar{V}_1 - \bar{V}_s}{R}$$

$$Z_1 = \frac{\bar{V}_1}{-\frac{2\bar{V}_1 - \bar{V}_s}{R}} = -\frac{\bar{V}_1 \cdot R}{2\bar{V}_1 - \bar{V}_s} = -\frac{(\bar{V}_s - \bar{I}_s R) R}{2(\bar{V}_s - \bar{I}_s R) - \bar{V}_s}$$

$$= -\frac{(\bar{V}_s - \bar{I}_s R) R}{\bar{V}_s - 2R}$$

Note: $\frac{\bar{V}_s}{\bar{I}_s} = \frac{10[\text{V}] / \underline{-30^\circ}}{0.1[\text{A}] / \underline{-120^\circ}} = 100[\Omega] / \underline{90^\circ}$

$$Z_1 = -\frac{(100[\Omega] / \underline{90^\circ} - 5[\Omega]) 5[\Omega]}{100[\Omega] / \underline{90^\circ} - 10[\Omega]}$$

$$= 4.98[\Omega] / \underline{177.2^\circ} = (-4.98 + j0.25)[\Omega]$$

not valid impedance

$$Z_1 = \underline{(-4.98 + j0.25)[\Omega]}$$

$$j\omega L = j0.25$$

$$L = \frac{0.25}{\omega} = \frac{0.25}{1000}$$

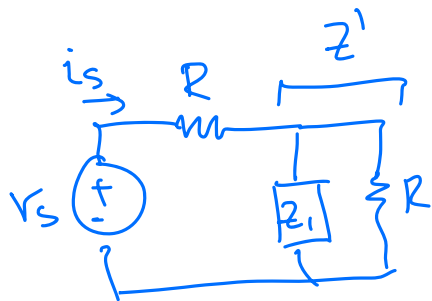
$$= 0.25[\text{mH}]$$

Equivalent Circuit:

$$-4.98[\Omega] \quad 0.25[\text{mH}]$$



not a valid
resistance



$$\bar{I}_s = \frac{\bar{V}_s}{R + Z'}$$

$$Z' = \frac{\bar{V}_s}{\bar{I}_s} - R = \frac{10[V] \angle -30^\circ}{0.15[A] \angle -120^\circ} - 5[\Omega]$$

$$= 100[\Omega] \angle +90^\circ - 5[\Omega] = (-5 + j100)[\Omega]$$

$$Z' = \frac{R Z_1}{R + Z_1}$$

$$R + Z_1 = \frac{R Z_1}{Z'}$$

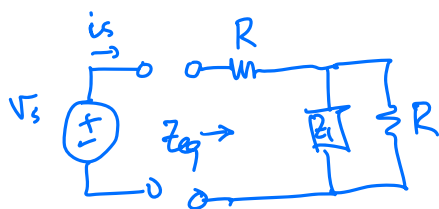
$$Z_1 \left(1 - \frac{R}{Z'}\right) = -R$$

$$Z_1 = \frac{-R}{1 - \frac{R}{Z'}} = \frac{-R Z'}{Z' - R}$$

$$= \frac{-5[\Omega](-5 + j100)[\Omega]}{(-5 + j100 - 5)[\Omega]}$$

$$= \frac{-5[\Omega](-5 + j100)}{-10 + j100}$$

$$= (-4.98 + j0.248)[\Omega]$$



Think of V_s as a test source, so that

$$Z_{eq} = \frac{\bar{V}_s}{\bar{I}_s} = R + Z_1 \parallel R$$

$$Z_1 \parallel R = \frac{\bar{V}_s}{\bar{I}_s} - R$$

$$\frac{Z_1 R}{Z_1 + R} = \frac{\bar{V}_s}{\bar{I}_s} - R$$

$$\frac{Z_1 R}{\frac{\bar{V}_s}{\bar{I}_s} - R} = Z_1 + R$$

$$-R = Z_1 \left(1 - \frac{R}{\frac{\bar{V}_s}{\bar{I}_s} - R}\right) = Z_1 \left(\frac{\frac{\bar{V}_s}{\bar{I}_s} - 2R}{\frac{\bar{V}_s}{\bar{I}_s} - R}\right)$$

$$Z_1 = \frac{-R \left(\frac{\bar{V}_s}{\bar{I}_s} - R\right)}{\frac{\bar{V}_s}{\bar{I}_s} - 2R}$$